Auteur: Rob Willemsen
Studierichting: Informatica
Oefeningenreeks 1B nr. 43.14

$$\Delta = (-(2+5i))^2 - 4 \cdot i \cdot (-10+8i)$$

$$\Delta = (4+20i-25) - (-40i-32) = -21+20i+32+40i = 11+60i$$

$$\begin{cases} x^2 - y^2 = 11 \\ 2xy = 60 \Rightarrow x^2(-y^2) = -900 \end{cases}$$

$$\lambda^2 - 11\lambda - 900 = 0$$

$$\Delta_\lambda = (-11)^2 - 4 \cdot 1 \cdot (-900) = 3721 = 61^2$$

$$\lambda_1 = \frac{11+61}{2} = 36 \Rightarrow x = \pm 6$$

$$\lambda_2 = \frac{11-61}{2} = -25 \Rightarrow y = \pm 5$$

$$\sqrt{\Delta} = 6+5i$$

$$z = \frac{2+5i \pm (6+5i)}{2i}$$

$$z_1 = \frac{8+10i}{2i} = \frac{4+5i}{i} = \frac{(4+5i)\cdot(-i)}{-i^2} = 5-4i$$

$$z_2 = \frac{-4}{2i} = \frac{-2}{i} = \frac{-2\cdot(-i)}{-i^2} = 2i$$

Antwoord: $z \in \{5-4i, 2i\}$

References